

Total No. of Questions : 10]

SEAT No. :

P6528

[Total No. of Pages : 2

[5870]-2009

T.E. (Computer Engineering)

DATABASE MANAGEMENT SYSTEM APPLICATIONS

(2012 Pattern) (Semester - I) (310244)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data, if necessary.

- Q1)** a) Compare SQL and NOSQL Databases with suitable examples of each? **[5]**
b) Define the term Normalization. Explain how 3NF treats the functional dependencies? **[5]**

OR

- Q2)** a) Elaborate query optimization with suitable example? **[5]**
b) Explain 1NF and 2NF in normalization. Give suitable example of each? **[5]**

- Q3)** a) Create the University Database having following specifications: **[5]**

- i) Department table (dept_no, dept_name, building_name)
Apply Unique constraint on name field.
- ii) Instructor table (ins_id, ins_name, dept_no, salary, mob_no)
Apply NOT NULL constraint on name field.
- iii) Course table (course id, title, dept_no, credits)
- iv) Teaches table (teacher id, course id, semester, year)

Create tables as per requirements (Mandatory)

Write the following queries with respect to above relations. (any 2)

- i) Increase the size of the title field of course relation.
 - ii) Create a view by showing a instructor name with a department name and its salary.
 - iii) Update salary of particular instructor using update view.
 - iv) Display details of employee having second highest salary.
- b) Describe cloud sourcing with respect to database? **[5]**

OR

P.T.O.

- Q4)** a) What are the advantages of allowing multiple transaction to run concurrently? [5]
 b) BASE Transactions ensures the properties like Basically Available, Soft State, Eventual Consistency. What is soft state of any system, how it is depend on Eventual consistency property? [5]
- Q5)** a) Explain concept of distributed Database. How distributed Transaction Works? [8]
 b) What is the concept of Parallel databases and Distributed databases? Explain in detail? [8]
- OR
- Q6)** a) Write a program in java/php/python to demonstrate the database connectivity for MySQL/oracle database? [8]
 b) Explain Speedup and Scale up in Parallel Databases. Explain with respect to Parallel Databases Operations. [8]
- Q7)** a) Write a short note on XML and its uses. Explain Querying in XML? [8]
 b) Explain different components of Hadoop in detail? [9]
- OR
- Q8)** a) Write XML Document for Book Data (Category, Title, Author, and Price). Write XQuery to retrieve all book information with price > 30. [8]
 b) Write short note on: [9]
 i) JSON
 ii) R programming
- Q9)** a) Explain supervised and unsupervised techniques in Machine Learning? Also state and explain different supervised and unsupervised ML algorithms? [8]
 b) Write a short note on following? [9]
 i) Operational Database and Data warehouse.
 ii) Map Reduce in Hadoop.
- OR
- Q10)** a) Write a short note clustering applications in Data mining? Explain K-means clustering algorithm? [8]
 b) Explain the need of ETL Process (Extract-Transform-Load) in data warehouse building? [9]

